

## Project Design Phase-II Technology Stack (Architecture & Stack)

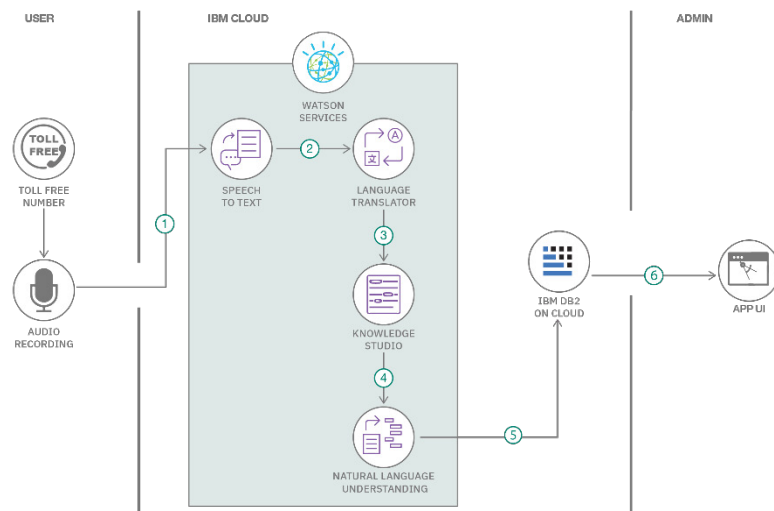
|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 20 Feb 2026                                                  |
| Team ID       | LTVIP2026TMIDS79394                                          |
| Project Name  | Toy craft tales: tableau's vision into toy manufacturer data |
| Maximum Marks | 4 Marks                                                      |

### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

### Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



### Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

| S.No | Component | Description | Technology |
|------|-----------|-------------|------------|
|------|-----------|-------------|------------|

|     |                                 |                                                                                                            |                                                                  |
|-----|---------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1.  | User Interface                  | How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.                                  | HTML, CSS, JavaScript / Angular Js / React Js etc. Java / Python |
| 2.  | Application Logic-1             | Logic for a process in the application                                                                     |                                                                  |
| 3.  | Application Logic-2             | Logic for a process in the application                                                                     | IBM Watson STT service                                           |
| 4.  | Application Logic-3             | Logic for a process in the application                                                                     | IBM Watson Assistant                                             |
| 5.  | Database                        | Data Type, Configurations etc.                                                                             | MySQL, NoSQL, etc.                                               |
| 6.  | Cloud Database                  | Database Service on Cloud                                                                                  | IBM DB2, IBM Cloudant etc.                                       |
| 7.  | File Storage                    | File storage requirements                                                                                  | IBM Block Storage or Other Storage Service or Local Filesystem   |
| 8.  | External API-1                  | Purpose of External API used in the application                                                            | IBM Weather API, etc.                                            |
| 9.  | External API-2                  | Purpose of External API used in the application                                                            | Aadhar API, etc.                                                 |
| 10. | Machine Learning Model          | Purpose of Machine Learning Model                                                                          | Object Recognition Model, etc.                                   |
| 11. | Infrastructure (Server / Cloud) | Application Deployment on Local System / Cloud<br>Local Server Configuration: Cloud Server Configuration : | Local, Cloud Foundry, Kubernetes, etc.                           |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                    | Technology                                          |
|------|--------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1.   | Open-Source Frameworks   | List the open-source frameworks used                                                           | Technology of Opensource framework                  |
| 2.   | Security Implementations | List all the security / access controls implemented, use of firewalls etc.                     | e.g. SHA-256, Encryptions, IAM Controls, OWASP etc. |
| 3.   | Scalable Architecture    | Justify the scalability of architecture (3 – tier, Micro-services)                             | Technology used                                     |
| 4.   | Availability             | Justify the availability of application (e.g. use of load balancers, distributed servers etc.) | Technology used                                     |

| S.No | Characteristics | Description                                                                                                               | Technology      |
|------|-----------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| 5.   | Performance     | Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc. | Technology used |

#### References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>